

Fault History, Case Scenarios and Diagnostic Evidence

Robot Weld 01 (robot-weld-01) | Sensor 1: AXIS_4_SERVO_TORQUE

Machine	Robot Weld 01 (machine ID robot-weld-01, six-axis spot-welding robot, Body shop, site vit-lab)
Sensor	Sensor 1: AXIS_4_SERVO_TORQUE, "Axis 4 servo torque", sensor ID ROBOT-WELD-01.AXIS_4_SERVO_TORQUE, type torque, unit Nm
Component / subsystem	Robot forearm (axis 4): axis 4 servo drive and gearbox; planned agent component tag robot_forearm_axis_4; suggested twin mesh 105-six-axis-welding-robot/arm-swing
Document title	Fault History, Case Scenarios and Diagnostic Evidence
Document ID	RW01-S1-DOC-05
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Purpose	Provides case-based diagnostic knowledge: worked scenarios that show how Sensor 1 (axis 4 servo torque) patterns correspond to fault situations, going from telemetry and context through abnormality, candidate faults, evidence, diagnosis, action and verification, so that a retrieval system can return a relevant case as supporting evidence.
Scope	Sixteen cases for Sensor 1 of Robot Weld 01 plus comparison tables. Cases C01 to C04 and C16 use the project fault GEARBOX_WEAR and the repository equations [P][D]; the other cases are illustrative engineering scenarios [E][I]. No real plant incident exists in the repository.

Provenance tags used throughout this document

[P]	Project value: taken from the repository or CONTEXT_SENSOR1.md (catalog limits, simulation equations, topics, tests).
[D]	Derived value: computed by the document authors from the repository's own simulation equations (noise-free unless stated).
[E]	Engineering reference: general industrial practice for robot servo-axis diagnostics. Not a manufacturer value and not taken from the repository.
[I]	Illustrative / proposed prototype value: chosen by the authors for this knowledge base and meant to be refined with real plant data.

Important: the telemetry for this machine is simulated. An ESP32 gateway (or a software simulator) plays the robot controller and generates values from documented equations. The limits in this document that are tagged [P] are illustrative constants chosen for the demo in the repository, not OEM specifications. Values tagged [E] and [I] are not manufacturer-certified.

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1. Fault History and Evidence Status

1.1 What fault history exists for this machine

The repository contains no recorded incidents, work orders, failure dates, repair records or field case studies for Robot Weld 01 [P]. Stored telemetry expires after 48 hours in the platform, so no long history is retained either. Every case in this document is therefore a simulated or illustrative scenario and must never be cited as plant history. The cases of the project fault are reproducible: the simulator is seeded and can be accelerated, so a run can be regenerated.

Table 1. Fault history record for Robot Weld 01

Item	Status	Note
Recorded failures of axis 4	None in repository	[P]
Work orders, repairs	None in repository	[P]
Field case studies	None in repository	[P]
Simulated fault runs	GEARBOX_WEAR (ramp 5 to 600 s); NORMAL	Reproducible with seed and speed options
Long-term telemetry	Not retained (48 h TTL)	Export evidence if needed
Planned evaluation	Root-cause accuracy, detection lead time, alerts collapsed, citation faithfulness, grounding, needs_binding rate	Gearbox wear is one of four evaluated scenarios [P]

1.2 How cases were built

- Project cases (C01, C02, C03, C16): the torque and TCP equations of the repository were replicated with a seeded random generator in 40 to 136 s runs [D]. The numbers are consistent with the equations; they differ in detail from any run of the repository's own generator because the random sequence differs.
- Illustrative cases (C04 to C15): numerical features are chosen to be consistent with the engineering signatures of Document 02 [E][I].
- Reproduction of project cases: `node server/scripts/simulate-devices.mjs --machines robot-weld-01 --scenario GEARBOX_WEAR --ramp 120 (options --speed, --seed, --crash) or inject GEARBOX_WEAR from the Faults tab with ENABLE_DEVICE_COMMANDS=true; choose Normal to reset [P].`
- Ground truth (the active scenario and ramp) is hidden from the agent; each case reasons from channel evidence only [P].

Table 2. How to read a case

Field	Meaning
Telemetry window	Samples used; 2 Hz; features computed as in Document 03
S	Severity index (M - 13.0) / 10.5
Level	L0 to L4, LS, as in Document 01
Score	Evidence points from Document 03 matrix
Basis	[P][D] project and derived; [E][I] illustrative

2. Case Summaries

Table 3. Case summary table

Case	Type	Basis	Sensor 1 pattern in one line	Level	Final diagnosis
C01	Gradual degradation	[P][D]	Ramp: M 13.8 to 23.2 Nm over 120 s, A6 up, TCP up	L0 to L3	F01 GEARBOX_WEAR
C02	Early-stage fault	[P][D]	M 14.6, A6 0.62, no limit crossed	L1	F01 incipient
C03	Sustained threshold violation	[P][D]	M 19.3, peak 27.1 Nm, warn in 9 of 10 cycles	L2	F01 stage 3
C04	Sudden abnormal spike	[E][I]	One sample 41.3 Nm; TCP step +0.25 mm	L4	F07 collision
C05	Intermittent abnormality	[E][I]	Spikes of 5 Nm at 2.5 s and 6.0 s in 60 percent of cycles	L1 (watch)	F05 backlash
C06	False alarm after program change	[E][I]	Warn flicker with M 15.8, swing +23 percent, A6 0.12	L2 on platform, not a fault	F10 program change
C07	Non-fault: cold-start warm-up	[E][I]	M 14.4 falling to 13.2 in 20 min	L1 transient	Warm-up effect (F09 / F02 component)
C08	Sensor fault: flatline	[E][I]	13.0 Nm constant, sigma 0	LS	F11 signal fault
C09	Measurement issue: data loss hides peaks	[E][I]	35 percent seq gaps, peaks missing	LS	F12 data path
C10	Friction family (lubrication)	[E][I]	M 17.0, A6 0.12, idle +1.1 Nm, warm-up offset	L2 (stage 2)	F02 lubrication loss
C11	Brake drag	[E][I]	M 16.2, idle +3.2 Nm, brake hot	L2	F03 brake drag
C12	External drag	[E][I]	Bump +3.4 Nm at 1.2 s only	L1	F06 dress pack
C13	Look-alike pair	[E][I]	Two windows with M 18 Nm: one wear, one friction	L2 both	F01 versus F02
C14	Transition F02 to F01	[E][I]	After relubrication M falls but A6 0.7 persists	L1	Emerging F01
C15	Post-repair verification	[E][I]	M 13.04, A6 0.10 after gearbox replacement	L0	Verified repair
C16	Severe / critical	[P][D][I]	M 24.1, sample 35.2 Nm, TCP 0.70 mm	L4	F01 beyond model

3. Detailed Cases: Project Fault GEARBOX_WEAR

3.1 C01 Gradual degradation under the default 120 s ramp [P][D]

Table 4. C01 case record

Element	Content
1. Case ID	C01. Simulated GEARBOX_WEAR injected at t = 0 with rampSec = 120. Replicated run, seeded.
2. Machine operating condition	Production cycle at 100 percent speed, warmed-up, program unchanged, payload unchanged; gun normal.
3. Sensor reading / time-series pattern	Eight 16 s windows (2 cycles each). See the telemetry table below: M rises from 13.8 to 23.2 Nm, per-cycle peak from 22.0 to 31.1 Nm, A6 from about 0.5 to 2.9 Nm, TCP mean from 0.14 to 0.65 mm.
4. Baseline comparison	Healthy baseline M 13.0, peak 21.7, A6 below 0.2, sigma 0.45, TCP 0.104. By window 3 (t = 32 s) M is 3.6 Nm above baseline (S 0.34) and A6 is 1.1 Nm.
5. Threshold comparison	First sample at or above 26 Nm occurs in window t = 48 to 64 s (peak 26.06 Nm). First sample at or above 30 Nm occurs in window t = 112 to 128 s (4 samples). Feature thresholds are crossed earlier: M above 15.0 at about t = 16 to 32 s; TCP above 0.25 at about t = 32 s.
6. Relevant operating state	Production cycle, steady, valid data (synced true, no gaps).
7. Symptoms	Rising TCP deviation, no mechanical alarm initially, warn then alarm on the twin (amber then red).
8. Candidate faults	F01 gearbox wear; F02 lubrication loss; F04 bearing; F05 looseness; F10 program change.
9. Evidence supporting each candidate	F01: mean rise, A6 growing in ratio 0.25 to 0.3 of dM, noise growth, TCP rise with ratio about 0.055 mm per Nm, swing unchanged, gun unchanged, ramp onset. F02: mean rise only. F04: noise rise. F05: TCP rise.
10. Evidence against alternatives	F02, F03, F09: A6 would be absent. F10, F07: swing would change and onset would be a step. F05: spikes would be at reversal phases, not smooth ripple. F11, F12: signal varies and other channels are normal.
11. Diagnostic reasoning	At t = 64 s the mean shift is 6.0 Nm, giving expected A6 about 1.7 Nm (observed 1.48), expected TCP shift about 0.31 mm (observed 0.34 mm) and swing unchanged. All ratios fall inside the accept ranges. The platform status remained normal until t about 50 s, but the feature evidence was available about 30 s earlier. Ramp onset excludes step causes.
12. Final diagnosis	GEARBOX_WEAR (F01): damaged gear mesh in axis 4 gearbox. Stage 2 at t = 32 s, stage 3 at t = 64 s, stage 4 at t = 112 s. Confidence high (14 points, margin 6 over F04).
13. Recommended inspection	At the first L1 (t about 16 s): lubricant sample, magnetic plug and trend request. At L2: gearbox oil, magnet, backlash, fasteners, noise (Document 04 tasks T1 and T2).
14. Corrective action	Replace gear set or gearbox if confirmed; lubricant and seals; re-master; recapture baseline.
15. Expected post-maintenance telemetry	M 12.5 to 13.5 Nm, A6 at most 0.2 Nm, sigma 0.40 to 0.55 Nm, peak at most 23.1 Nm, TCP at most 0.15 mm, status normal.
16. Verification result	Simulated: sending the NORMAL scenario returns the mean to 13.0 Nm within one cycle; ack shows scenario NORMAL (project behaviour).

Table 5. C01 telemetry by 16 s window (replicated run, ramp 120 s) [D]

Window start (s)	Mean r	M (Nm)	S	Per-cycle peak (Nm)	A6 (Nm)	TCP mean (mm)	Samples warn / alarm	Level
0	0.06	13.76	0.07	21.99	0.53 (floor)	0.14	0 / 0	L0 to L1
16	0.20	14.98	0.19	22.85	0.76	0.21	0 / 0	L1
32	0.33	16.58	0.34	24.17	1.11	0.29	0 / 0	L1 to L2 features
48	0.46	17.83	0.46	25.49	1.33	0.36	1 / 0	L2 begins
64	0.60	18.99	0.57	26.61	1.48	0.44	4 / 0	L2
80	0.73	20.65	0.73	28.46	1.82	0.51	7 / 0	L2
96	0.86	21.67	0.83	29.31	2.03	0.58	10 / 0	L2 to L3
112	0.98	23.19	0.97	31.07	2.93	0.65	7 / 4	L3 to L4 (M near 23.5)

Note: A6 over a 16 s window has a noise floor of about 0.15 to 0.4 Nm and the ramp changes r during the window, so early A6 values are not exact. The trend and the later windows are consistent with 3.0 r Nm [D].

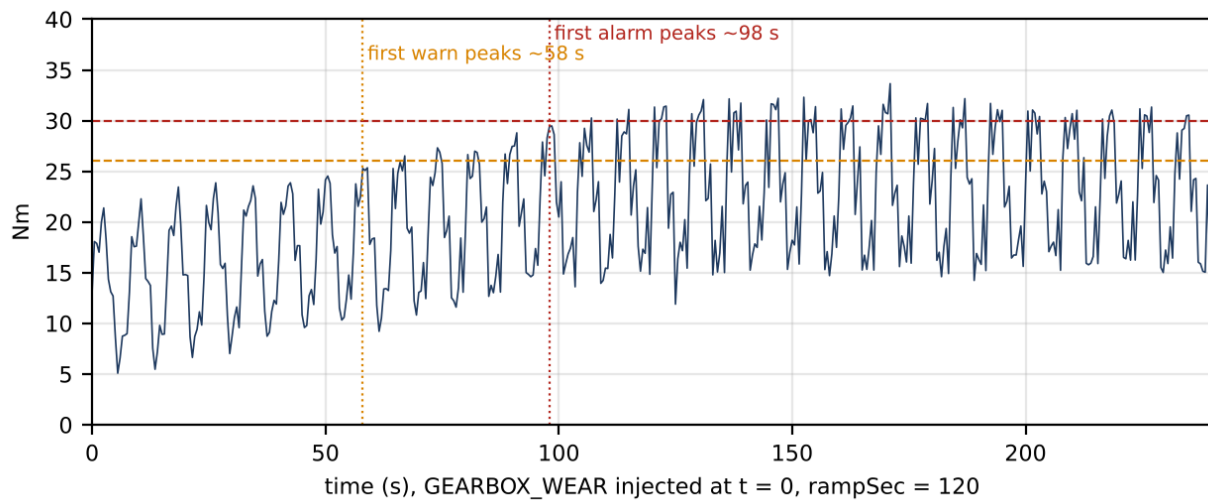


Figure 1. Replicated 240 s run: GEARBOX_WEAR injected at $t = 0$ with the default 120 s ramp; cycle peaks cross 26 Nm at about 58 s and 30 Nm at about 98 s [D].

3.2 C02 Early-stage fault below every platform limit [P][D]

Table 6. C02 case record

Element	Content
1. Case ID	C02. Simulated GEARBOX_WEAR held at severity $r = 0.15$, 40 s window (5 cycles), seeded.
2. Machine operating condition	Production cycle, steady, warmed up.
3. Sensor reading / time-series pattern	M 14.59 Nm, per-cycle peak mean 22.89 Nm (maximum sample 24.2 Nm), minimum mean 6.22 Nm, A6 0.62 Nm, sigma 0.55 Nm, TCP mean 0.19 mm.
4. Baseline comparison	dM +1.6 Nm (S 0.15), peak +1.2 Nm, minimum +1.9 Nm, A6 above 0.4 watch, sigma 0.55 at the watch boundary, TCP +0.086 mm.
5. Threshold comparison	No sample reaches 26 Nm; platform status normal throughout. Feature thresholds: M above 13.5 (watch), A6 above 0.4 (watch), peak below 23.5.
6. Relevant operating state	Production cycle, steady, valid data.
7. Symptoms	None visible to operators; TCP slightly above baseline.
8. Candidate faults	F01 incipient wear; F02 lubrication; F03 brake drag; F09 thermal; F10 minor program change.
9. Evidence supporting each candidate	F01: $A6 / dM = 0.39$, $(kN - 1) / dM = 0.14$, $dTCP / dM = 0.054$ mm per Nm, swing unchanged. F02 and F03 explain the mean only. F09 explains it if M is reversible.
10. Evidence against alternatives	F03: no idle offset recorded. F09: M not reversible with temperature. F10: no program change in log. A6 and TCP ratios are not expected from friction faults.
11. Diagnostic reasoning	The three locked ratios all fall inside the accept ranges even though the platform status is normal, so the evidence is specific rather than noise: M exceeds the healthy band by 10 sigma of the 5-cycle mean (0.05 Nm). A6 exceeds the 5-cycle healthy 95th percentile of 0.18 by a factor of 3.
12. Final diagnosis	Incipient GEARBOX_WEAR (F01, stage 1). Level L1 watch, severity low, confidence medium to high (score 14, but baseline not independently confirmed).
13. Recommended inspection	At next planned stop: lubricant sample and magnetic plug (T1); backlash baseline (T2); check lubrication records.
14. Corrective action	Trend each shift; relubricate if due per OEM; schedule gear inspection if M exceeds 15 Nm.
15. Expected post-maintenance telemetry	After successful service M returns to 12.5 to 13.5 Nm and A6 to at most 0.2 Nm; if A6 persists the wear is real.
16. Verification result	Platform shows no state change either before or after; verification by features only.

3.3 C03 Sustained threshold violation: persistent warning [P][D]

Table 7. C03 case record

Element	Content
1. Case ID	C03. Simulated GEARBOX_WEAR held at $r = 0.6$, 80 s (10 cycles), seeded.
2. Machine operating condition	Production cycle, steady, valid data.
3. Sensor reading / time-series pattern	M 19.31 Nm, per-cycle peak mean 27.12 Nm (maximum sample 27.9 Nm), minimum mean 11.4 Nm, A6 1.70 Nm, sigma 0.82 Nm, TCP mean 0.43 mm; 22 of 160 samples (13.8 percent) at or above 26 Nm; none at or above 30 Nm; 9 of 10 cycles contain a sample at or above 26 Nm.
4. Baseline comparison	dM +6.3 Nm (S 0.60), peak +5.4 Nm, TCP +0.33 mm.
5. Threshold comparison	Peak above warnHigh 26 for at least 2 consecutive cycles: WARNING opened (rule R-09). Below alarmHigh 30 Nm (maximum 27.9, margin 2.1 Nm).
6. Relevant operating state	Production cycle, steady.
7. Symptoms	Weld position scatter possible; amber on the twin each cycle.
8. Candidate faults	F01 stage 3; F02; F07 payload; F05 looseness.
9. Evidence supporting each candidate	F01: $A6 / dM = 0.27$, $(kN - 1) / dM = (0.82 / 0.45 - 1) / 6.3 = 0.13$, $dTCP / dM = 0.052$, swing unchanged. F02: only mean. F07: swing would change. F05: TCP shift per Nm would be much larger.
10. Evidence against alternatives	No program or payload change; no step in M; spikes absent.
11. Diagnostic reasoning	The platform flickers warn once per cycle because the peak crosses 26 Nm for 0.5 to 1 s per cycle. Persistence is satisfied on the peak envelope (9 of 10 cycles). With ratios in range, the leading diagnosis is F01.
12. Final diagnosis	GEARBOX_WEAR stage 3; level L2 warning; severity medium; confidence high.
13. Recommended inspection	Inspection at the next planned stop: oil, magnet, backlash, noise, fasteners.
14. Corrective action	Schedule gear replacement; reduce duty if time to alarm is short. With slope 0.0875 Nm/s the alarm-equivalent M of 21.6 Nm is 26 s away at this simulator rate; for a real drift of 0.1 Nm per day it is about 23 days [D][E].
15. Expected post-maintenance telemetry	M 12.5 to 13.5, A6 at most 0.2, peak at most 23.1.
16. Verification result	Platform shows normal for 600 s after repair.

3.4 C16 Severe and critical condition: gearbox wear beyond the modeled envelope [P][D][I]

Table 8. C16 case record

Element	Content
1. Case ID	C16. Simulated GEARBOX_WEAR held at $r = 1.0$ (40 s window) with an additional isolated sample of 35.2 Nm [I].
2. Machine operating condition	Production cycle, alarm active for 10 minutes.
3. Sensor reading / time-series pattern	M 23.54 Nm, per-cycle peak mean 32.51 Nm, maximum sample 33.7 Nm in the replicated window; one later sample 35.2 Nm; minimum mean 15.18 Nm; A6 3.34 Nm; sigma 1.13 Nm; TCP 0.66 mm; 10 samples warn and 17 alarm of 80 (12.5 and 21 percent).
4. Baseline comparison	dM +10.5 Nm (S 1.0), minimum +10.9 Nm, TCP +0.56 mm.
5. Threshold comparison	Per-cycle peak above 30 every cycle; M at 23.5 (critical-equivalent); sample at or above 34 Nm confirmed in the next cycle; alarm persistence longer than 10 minutes: CRITICAL by three independent criteria.
6. Relevant operating state	Production cycle at full speed.
7. Symptoms	Noise at axis 4, weld position error above tolerance, red pulsing mesh.
8. Candidate faults	F01 stage 5; F07 overload or collision damage; F04 bearing; F05 looseness.
9. Evidence supporting each candidate	F01: all ratios in range at full severity (A6 / dM 0.32, TCP / dM 0.053). F07 would explain 35.2 Nm only as a spike.
10. Evidence against alternatives	Gradual onset and locked ratios exclude F07 as the primary cause; F11 excluded because the cycle pattern is present.
11. Diagnostic reasoning	Every criterion for critical is met; the modeled fault saturates at a noise-free peak of 32.3 Nm, so a sample at 35.2 Nm is a tail event of full-severity noise (about 2.6 sigma above the peak) or a damage event; either way the machine is at the end of its modeled life.
12. Final diagnosis	F01 beyond model, critical; confidence high; treat as probable tooth or bearing damage until inspection says otherwise.
13. Recommended inspection	Stop at a safe point; lock-out; gearbox opening by specialist; inspect teeth, bearings, shaft.
14. Corrective action	Replace gearbox; inspect motor and brake; re-master; verification run.
15. Expected post-maintenance telemetry	After repair features return to the healthy band.
16. Verification result	Return to service only after 600 s normal status and TCP at most 0.15 mm.

4. Detailed Cases: Event, Intermittent and Non-Fault Conditions

4.1 C04 Sudden abnormal spike: collision [E][I]

Table 9. C04 case record

Element	Content
1. Case ID	C04. Illustrative collision of the weld gun with a fixture during one cycle of Robot Weld 01.
2. Machine operating condition	Production cycle at 100 percent speed, warmed up, program unchanged until the event; the robot stops through its safety function after the contact.
3. Sensor reading / time-series pattern	One sample of 41.3 Nm at 2.5 s of one cycle, with healthy values before and after it. In the following cycles after restart M is 13.8 Nm (+0.8), sigma 0.52 Nm, A6 0.14 Nm. TOOL_CENTER_POINT_DEVIATION mean steps from 0.10 to 0.35 mm within the same minute.
4. Baseline comparison	Spike 19.6 Nm above the healthy peak of 21.75 Nm; M +0.8 Nm; sigma +0.07 Nm; TCP +0.25 mm (dTCP / dM = 0.31 mm per Nm, six times the F01 ratio).
5. Threshold comparison	The sample exceeds alarmHigh 30 Nm and the proposed critical level 34 Nm. The per-cycle peak envelope falls back below 26 Nm in the next cycle, so no persistence-based alarm would open: the event is detected only by the single-sample rule.
6. Relevant operating state	Production, then safety stop and restart. Thresholds apply to the cycles before and after, not to the stop itself.
7. Symptoms	Controller collision or safety stop event; visible contact marks on the fixture or gun; weld position error afterwards.
8. Candidate faults	F07 collision; F11 isolated spike; F12 data glitch; F05 looseness released by the impact; F01 sudden tooth damage.
9. Evidence supporting each candidate	F07: isolated spike (3), TCP step large for the torque shift (1), other channel affected (1), step onset (3), unchanged swing and gun, event log shows collision (5): total 14. F05: TCP rise large for dM (3) plus baseline items: total 6. F11: isolated spike (2), step (3), gun unchanged (1): 6.
10. Evidence against alternatives	F01: onset is a step, not a ramp, and A6 is unchanged at 0.14 Nm. F11 and F12: seq is continuous, the device is online and TCP changes consistently with the spike.
11. Diagnostic reasoning	Rule R-11 alone would classify a single unconfirmed sample as a transient; rule R-27 overrides it because the spike is accompanied by a TCP step and an event log entry. The residual mean rise and noise increase after the event show that damage cannot be excluded, so the case stays at L4 until the mechanical inspection is complete.
12. Final diagnosis	F07 overload by collision, possible secondary damage to the axis 4 gear train or coupling. Level L4 critical until inspected; confidence high for the event, medium for the extent of damage.
13. Recommended inspection	Visual damage to tool and gun; tool data; fastener torque and torque marks; backlash at the flange; gearbox noise; mastering marks of axis 4 (Document 04 tasks T4 and T2).
14. Corrective action	Repair or replace damaged parts; replace the coupling or gearbox if play or debris is found; re-master axis 4; re-baseline Sensor 1 and TCP.
15. Expected post-maintenance telemetry	M 12.5 to 13.5 Nm, sigma 0.40 to 0.55 Nm, A6 at most 0.2 Nm, TCP mean at most 0.15 mm; no sample above 23.1 Nm.
16. Verification result	Illustrative pass: 600 s with no spike; TCP back to 0.10 mm; backlash within the OEM limit; baseline stored.

4.2 C05 Intermittent abnormality: backlash spikes at reversals [E][I]

Table 10. C05 case record

Element	Content
1. Case ID	C05. Illustrative worn coupling with free play between motor and gearbox of axis 4.
2. Machine operating condition	Production cycle at 100 percent speed, steady; condition developed over several weeks.
3. Sensor reading / time-series pattern	M 13.3 Nm; A6 0.35 Nm; sigma 0.50 Nm. In about 60 percent of cycles a spike of 4 to 6 Nm appears at about 2.5 s and at about 6.0 s of the cycle, the phases at which the torque reverses direction of change. TCP mean 0.21 mm.
4. Baseline comparison	dM +0.3 Nm; dTCP +0.11 mm, so dTCP / dM = 0.37 mm per Nm, about seven times the F01 ratio of 0.052. No change of swing.
5. Threshold comparison	No sample at or above 26 Nm except in about 5 percent of cycles where the 2.5 s spike reaches 27 Nm, so the per-cycle envelope never satisfies the two-cycle persistence rule. Platform status is normal almost all the time.
6. Relevant operating state	Production, steady, valid data.
7. Symptoms	Audible clunk at direction changes, weld position scatter, repeatability complaints from the quality team.
8. Candidate faults	F05 backlash or looseness; F04 bearing degradation; F01 early wear; F06 dress pack drag.
9. Evidence supporting each candidate	F05: spikes locked to the reversal phases (4), TCP rise large for the torque shift (3), swing unchanged (1), gun unchanged (1), gradual onset (1): total 10. F04: sparse spikes and noise but not phase-locked: 4. F01: 4.
10. Evidence against alternatives	F01: dM is only +0.3 Nm and A6 is 0.35 Nm, below the 0.4 watch level and not in ratio with the TCP shift. F06: the disturbance is impulsive, not a smooth bump. F04: spikes are repeatable at fixed phases, not random.
11. Diagnostic reasoning	Intermittency arises because the free play is only crossed when the load torque reverses sign through zero under certain combinations of speed and temperature. Phase-locking to the torque reversals, together with a TCP shift that is far larger than the mean torque shift suggests, is the discriminator that separates looseness from wear.
12. Final diagnosis	F05 backlash or mechanical looseness; level L1 watch with quality risk; severity medium; confidence medium to high.
13. Recommended inspection	Fastener torque of gearbox and motor mounting, coupling and key condition, backlash at the flange (Document 04 tasks T4 and T2).
14. Corrective action	Retorque to the OEM specification; replace coupling or key if worn; adjust or replace the gear set if backlash exceeds the OEM limit; re-master axis 4 afterwards.
15. Expected post-maintenance telemetry	Spikes absent over 20 cycles; A6 at most 0.2 Nm; TCP mean about 0.10 mm; M 12.5 to 13.5 Nm.
16. Verification result	Illustrative pass: spike count zero over 20 cycles and 600 s with no warn; backlash measurement recorded as the new baseline.

4.3 C06 False alarm after a program change [E][I]

Table 11. C06 case record

Element	Content
1. Case ID	C06. Illustrative start of a new product program on Robot Weld 01, payload unchanged.
2. Machine operating condition	Production with a new program ID, speed override 100 percent, warm-up complete.
3. Sensor reading / time-series pattern	M steps from 13.0 to 15.8 Nm within one cycle; cycle swing grows to 8.6 Nm (from 7.0); per-cycle peak 26.4 Nm in several consecutive cycles; A6 0.12 Nm; sigma 0.47 Nm; TCP mean 0.10 mm. The waveform shape also differs from the stored template.
4. Baseline comparison	dM +2.8 Nm; swing +23 percent; A6 and TCP at baseline; sigma at baseline.
5. Threshold comparison	Per-cycle peak at or above 26 Nm for three consecutive cycles: a warning would open on thresholds alone (rule R-09), and the twin shows an amber flicker once per cycle.
6. Relevant operating state	Production, but with a changed program: the baseline of Document 01 does not apply to the new motion.
7. Symptoms	None on the machine; the operator knows the program changed.
8. Candidate faults	F10 program change; F07 payload or tool data change; F01 early wear; F02 lubrication.
9. Evidence supporting each candidate	F10: step onset (3), swing changed (3), other channels may change (1), event log shows planned program change (5), dM (1): total 13. F07: swing changed (3), step (3), dM (2), other (1), event log (2): 11.
10. Evidence against alternatives	F01: A6 absent and TCP unchanged, swing changed (F01 leaves the swing unchanged). F02 and F03: swing would not change and no idle offset is reported.
11. Diagnostic reasoning	A step in mean with a change of swing and no ripple is the signature of a load or program change, not of wear (rule R-12 and R-26). The margin between F10 and F07 is 2 points, so the agent must confirm from the tool data that the payload did not change before closing the case as a program change.
12. Final diagnosis	F10 program change: not a fault. The threshold alert is a false alarm for gearbox wear. Confidence medium until tool and payload data are confirmed.
13. Recommended inspection	Check the program log and tool and payload data only; no mechanical inspection.
14. Corrective action	Re-baseline Sensor 1 for the new program and store it with the program ID; adjust the watch levels if the new program is permanent.
15. Expected post-maintenance telemetry	New baseline stable over 20 cycles; A6 at most 0.2 Nm; sigma 0.40 to 0.55 Nm.
16. Verification result	Illustrative result: the alert is closed with the note program change; no wear conclusion; the old baseline is kept for the old program.

4.4 C07 Non-fault: cold-start warm-up [E][I]

Table 12. C07 case record

Element	Content
1. Case ID	C07. Illustrative restart after a weekend stop, gearbox at ambient temperature.
2. Machine operating condition	First 20 minutes of production after the restart, program unchanged.
3. Sensor reading / time-series pattern	M is 14.4 Nm in the first cycles and falls steadily to 13.2 Nm after 20 minutes; A6 stays at 0.10 Nm; sigma 0.47 Nm; TCP 0.10 mm throughout.
4. Baseline comparison	dM starts at +1.4 Nm and decays to +0.2 Nm; no change in A6, sigma or TCP.
5. Threshold comparison	M is above the 13.5 Nm watch level only during the first minutes; no sample reaches 26 Nm; the per-cycle peak stays below 23.5 Nm.
6. Relevant operating state	Warm-up after a long stop. Thresholds and baselines must not be evaluated in this state (rule R-06).
7. Symptoms	None.
8. Candidate faults	F09 cold viscosity; F02 lubrication degradation; F01 incipient wear.
9. Evidence supporting each candidate	F09 and the warm-up component of F02: the offset is reversible and decays with running time.
10. Evidence against alternatives	F01: A6 and TCP are unchanged and wear does not decay after restart. F02 persistent part: the offset disappears fully.
11. Diagnostic reasoning	A reversible offset with a time constant of minutes, no ripple and no TCP effect is explained by lubricant viscosity. The state gate applies first; the agent should not open an investigation. If the offset does not decay, the case becomes C10.
12. Final diagnosis	Warm-up effect (normal operation). No fault; level L1 only during the first minutes.
13. Recommended inspection	None.
14. Corrective action	None. Exclude the warm-up cycles from baseline and trend calculations.
15. Expected post-maintenance telemetry	M within 12.5 to 13.5 Nm after warm-up.
16. Verification result	Illustrative result: no alert; the stable M after 20 minutes confirms that the effect is thermal.

4.5 C08 Sensor fault: flatline [E][I]

Table 13. C08 case record

Element	Content
1. Case ID	C08. Illustrative controller interface that stops updating the torque value while the robot continues to move.
2. Machine operating condition	Program running, robot moving normally, no controller alarms.
3. Sensor reading / time-series pattern	Torque constant at 13.0 Nm for 60 s; sigma 0.0 Nm; TCP deviation shows its normal 0.008 mm noise around 0.10 mm; gun temperature follows its weld cycle.
4. Baseline comparison	Variance zero versus 0.45 Nm; swing 0 versus 7.0 Nm; no 8 s periodicity.
5. Threshold comparison	Status normal; no limit is crossed, so a threshold-only agent would report a healthy axis.
6. Relevant operating state	Production (the program is running).
7. Symptoms	The twin shows a steady colour and no cycle trace; operators see nothing abnormal at the robot.
8. Candidate faults	F11 flatline; F12 stale data; idle state; F10 program change.
9. Evidence supporting each candidate	F11: flatline (4), step onset (3), gun unchanged (1), and rule R-19 for sigma below 0.1 Nm (4): total 12. The other channels move normally and the program is running.
10. Evidence against alternatives	F12: seq is continuous, the device is not stale and the other channels update. Idle: the program is running. F10: the swing would still be present in a changed program.
11. Diagnostic reasoning	A healthy axis shows sigma 0.45 Nm and a swing of 7 Nm; a perfectly constant value is physically impossible while the arm moves. The state of the other channels excludes a data-path failure.
12. Final diagnosis	F11 signal fault, level LS; no mechanical conclusion.
13. Recommended inspection	Data-path validation (Document 04 task T6): controller interface, scaling, gateway diag fields.
14. Corrective action	Correct the interface or scaling; restart the gateway or controller task if needed.
15. Expected post-maintenance telemetry	Cycle trace returns with sigma about 0.45 Nm and the 8 s periodicity.
16. Verification result	Illustrative pass: autocorrelation peak at 8 s restored; features within baseline.

4.6 C09 Measurement issue: data loss hides peaks [E][I]

Table 14. C09 case record

Element	Content
1. Case ID	C09. Illustrative weak Wi-Fi link on the ESP32 gateway while real wear is developing.
2. Machine operating condition	Production cycle; underlying wear at S about 0.55.
3. Sensor reading / time-series pattern	35 percent of the samples are missing; the observed per-cycle peak is 24.8 Nm while the true peak is about 26.5 Nm; seq has gaps; diag shows rising wifiReconnects and a low rssi.
4. Baseline comparison	The observed peak underestimates the true peak by about 1.7 Nm; the observed M is stable but noisy.
5. Threshold comparison	No sample at or above 26 Nm is seen in 4 of 5 cycles: the platform shows normal although the true warn limit is crossed in about half of the cycles.
6. Relevant operating state	Production, steady.
7. Symptoms	Gaps in the twin trace and in the history chart.
8. Candidate faults	F12 data-path fault; masked F01 wear.
9. Evidence supporting each candidate	F12: gaps in seq, reconnect counters rising, low rssi. F01: the surviving samples still show a mean shift and TCP rise.
10. Evidence against alternatives	F11: the signal varies and other channels also have gaps.
11. Diagnostic reasoning	Missing samples can remove the one or two samples near the cycle peak, which are the only ones that cross the limit, so a data-loss window under-reports severity. Rule R-02 marks the window incomplete and lowers the confidence; the mean, TCP and A6 of the surviving samples must still be evaluated.
12. Final diagnosis	F12 data-path fault with a possible masked F01; confidence medium.
13. Recommended inspection	Network and gateway check (rssi, reconnects, band); then a complete window.
14. Corrective action	Improve the Wi-Fi link (2.4 GHz, signal strength), restart the gateway, and re-evaluate with a complete window.
15. Expected post-maintenance telemetry	No seq gaps; peak and features reflect the true state.
16. Verification result	Illustrative result: with a complete window the true peak of 26.5 Nm is seen and the case becomes a C03-type warning.

5. Detailed Cases: Friction, Brake, Drag and Transitions

5.1 C10 Friction family: lubrication loss [E][I]

Table 15. C10 case record

Element	Content
1. Case ID	C10. Illustrative overdue relubrication of the axis 4 gearbox.
2. Machine operating condition	Production, steady, after warm-up; the lubrication record shows the last service is beyond the OEM schedule.
3. Sensor reading / time-series pattern	M 17.0 Nm (S 0.38); A6 0.12 Nm; sigma 0.52 Nm; TCP 0.11 mm; idle torque +1.1 Nm above the earlier record; M is 1.0 Nm higher right after a cold start.
4. Baseline comparison	dM +4.0 Nm; A6 / dM = 0.03 (F01 expects 0.29); dTCP / dM = 0.002 (F01 expects 0.052).
5. Threshold comparison	Per-cycle peak mean 24.4 Nm; no sample at or above 26 Nm; level L1 to L2 by features, platform status normal.
6. Relevant operating state	Production.
7. Symptoms	Warm gearbox housing, grease discolouration at the vent.
8. Candidate faults	F02 lubrication loss; F03 brake drag; F09 thermal; F01 wear.
9. Evidence supporting each candidate	F02: gradual onset (1), idle offset (1), reversible part (1), swing and gun unchanged (2), dM (2): 7, with the lubrication record as extra context. F03 and F09 also score 7 on the matrix.
10. Evidence against alternatives	F01: A6 and TCP ratios absent. F03: the brake is cold and the idle offset is smaller than the mean shift. F09: only part of the offset is reversible.
11. Diagnostic reasoning	The mean shift without ripple or TCP effect indicates friction. Three friction candidates tie, so the decisive tests are ordered by cost: lubricant check (cheapest and supported by the overdue record), then warm-up comparison, then brake voltage.
12. Final diagnosis	F02 lubrication degradation; level L2 by features; severity medium; confidence low until the lubricant is inspected, then medium to high.
13. Recommended inspection	Lubricant level, colour and sample; magnetic plug; seals (Document 04 task T1).
14. Corrective action	Relubricate or replace the lubricant per the OEM grade; replace seals if leaking; run a warm-up and re-baseline.
15. Expected post-maintenance telemetry	M 12.5 to 13.5 Nm; warm-up offset at most 0.5 Nm; A6 at most 0.2 Nm.
16. Verification result	Illustrative pass: warm-up offset 0.3 Nm after service; features healthy; no particles in the sample.

5.2 C11 Brake drag [E][I]

Table 16. C11 case record

Element	Content
1. Case ID	C11. Illustrative holding brake that releases late and drags during motion.
2. Machine operating condition	Production cycle; condition appears after a power event.
3. Sensor reading / time-series pattern	M 16.2 Nm; A6 0.1 Nm; sigma 0.47 Nm; idle torque +3.2 Nm above the earlier record; brake housing hotter than neighbouring axes; TCP 0.12 mm.
4. Baseline comparison	dM +3.2 Nm and idle offset +3.2 Nm: equal within measurement error.
5. Threshold comparison	No sample at or above 26 Nm; level L1 to L2 by features.
6. Relevant operating state	Production and idle both show the offset.
7. Symptoms	Brake release click absent, warm housing, higher drive current at standstill.
8. Candidate faults	F03 brake drag; F08 drive offset; F02 lubrication loss.
9. Evidence supporting each candidate	F03: idle offset (3), brake locally hot (3), step onset (2), dM (2), swing and gun unchanged (2): total 12. F08: idle offset (3), step (2), dM (1) and others: 8.
10. Evidence against alternatives	F08: the brake is hot, which a pure measurement offset would not cause. F02: idle offset would be smaller than the mean shift and the housing would not be hot.
11. Diagnostic reasoning	A constant Coulomb torque from a dragging brake adds the same torque during motion and at idle, so the equal offsets are the discriminator. The temperature rise confirms energy dissipation in the brake.
12. Final diagnosis	F03 brake drag; level L2; severity medium to high because overheating can follow.
13. Recommended inspection	Brake supply voltage during release, release timing, disc and armature (Document 04 task T3).
14. Corrective action	Correct the supply voltage or wiring; replace the brake if worn; verify release timing in the controller.
15. Expected post-maintenance telemetry	Idle offset zero; M 12.5 to 13.5 Nm; brake temperature normal.
16. Verification result	Illustrative pass: idle torque back to the earlier hold value; 600 s with normal status.

5.3 C12 External drag: dress pack bump [E][I]

Table 17. C12 case record

Element	Content
1. Case ID	C12. Illustrative tight dress pack after a tool change.
2. Machine operating condition	Production cycle, steady, program unchanged.
3. Sensor reading / time-series pattern	A bump of +3.4 Nm appears at about 1.2 s in every cycle; M 13.6 Nm; A6 0.2 Nm; sigma 0.46 Nm; TCP 0.10 mm.
4. Baseline comparison	Residual to the template is 3.4 Nm in one window about 1 s wide and below 0.5 Nm elsewhere.
5. Threshold comparison	No limit is crossed: the peak stays below 23.5 Nm because the bump lies on the rising part of the cycle, not at the maximum.
6. Relevant operating state	Production, steady.
7. Symptoms	Visible cable tension at one pose; jerky motion at that pose.
8. Candidate faults	F06 external drag; F05 looseness; F01 early wear.
9. Evidence supporting each candidate	F06: smooth bump in one phase window (4), swing and gun unchanged (2), step onset after the tool change (1): total 7. F05: 4. F07: 5.
10. Evidence against alternatives	F05: the disturbance is smooth, not an impulse at a reversal. F01: the shift is local, not whole-cycle, and A6 is below 0.4 Nm.
11. Diagnostic reasoning	Posture-dependent drag affects only the segment of the stroke where the cable is tensioned; its position in the cycle tells which pose.
12. Final diagnosis	F06 external drag; level L1; severity low to medium (cable damage risk).
13. Recommended inspection	Dress pack routing and slack across the whole stroke; seal and cover contact (Document 04 task T5).
14. Corrective action	Re-route the dress pack, adjust slack, replace worn guides, clean spatter.
15. Expected post-maintenance telemetry	Bump removed; M 12.5 to 13.5 Nm.
16. Verification result	Illustrative pass: residual below 1 Nm in every phase of the cycle.

5.4 C13 Similar patterns, different causes: wear versus friction [E][I]

Table 18. C13 case record

Element	Content
1. Case ID	C13. Two windows from two periods on the same robot: W-A (gearbox wear) and W-B (lubrication loss).
2. Machine operating condition	Both in production, steady, warmed up (W-B also analysed cold).
3. Sensor reading / time-series pattern	W-A: M 18.1, peak 26.3, A6 1.6, sigma 0.78, TCP 0.41 mm, idle offset 0. W-B: M 18.0, peak 26.2, A6 0.10, sigma 0.55, TCP 0.11 mm, idle +1.4 Nm, M falls by 1.2 Nm after warm-up.
4. Baseline comparison	Both have dM about +5.0 Nm (S about 0.48).
5. Threshold comparison	Both are at the warning boundary: peaks 26.2 and 26.3 Nm.
6. Relevant operating state	Production, steady.
7. Symptoms	W-A: TCP drift and weld position scatter. W-B: warm housing.
8. Candidate faults	F01 versus F02, with F03 and F09 as further friction candidates.
9. Evidence supporting each candidate	W-A: A6 / dM 0.32 and dTCP / dM 0.061 within the F01 ranges, kN 1.73. W-B: A6 / dM 0.02, dTCP / dM 0.001, reversible warm-up part, idle offset.
10. Evidence against alternatives	W-A excludes F02 because ripple and TCP rise are present; W-B excludes F01 because they are absent.
11. Diagnostic reasoning	The mean and the peak are identical, so a threshold-only view cannot separate them. The ratios A6 / dM and dTCP / dM, the idle offset and the warm-up behaviour separate them reliably.
12. Final diagnosis	W-A is F01 gearbox wear (stage 3); W-B is F02 lubrication loss (stage 2, friction family).
13. Recommended inspection	W-A: gearbox oil, magnet, backlash. W-B: lubricant level and condition.
14. Corrective action	W-A: plan gearbox replacement. W-B: relubricate.
15. Expected post-maintenance telemetry	Each returns to baseline after its own repair; W-A also needs the TCP back at 0.10 mm.
16. Verification result	Illustrative pass for both; applying the wrong repair would leave the other window unchanged.

5.5 C14 Transition from lubrication loss to wear [E][I]

Table 19. C14 case record

Element	Content
1. Case ID	C14. Illustrative continuation of C10 after relubrication.
2. Machine operating condition	Production, steady, three days after the lubricant service.
3. Sensor reading / time-series pattern	After service M falls from 17.0 to 15.2 Nm but not to baseline; A6 rises from 0.12 to 0.70 Nm; TCP 0.17 mm; the oil sample taken during service showed fines on the magnet.
4. Baseline comparison	dM +2.2 Nm; A6 / dM = 0.32; dTCP / dM = 0.03.
5. Threshold comparison	No limit crossed; M above the 15.0 Nm early-degradation boundary.
6. Relevant operating state	Production.
7. Symptoms	None noticed.
8. Candidate faults	F01 emerging wear; residual F02; F04.
9. Evidence supporting each candidate	F01: with the friction part removed the remaining mean shift carries the ripple ratio of F01 (A6 / dM 0.32), swing unchanged. Matrix score F01 12, F04 6.
10. Evidence against alternatives	A purely friction fault would have returned to baseline after service.
11. Diagnostic reasoning	Removing the friction contribution exposes the underlying wear: the part of the shift that did not respond to lubrication is the part with ripple.
12. Final diagnosis	Emerging GEARBOX_WEAR (F01) stage 1 to 2.
13. Recommended inspection	Oil particle analysis and backlash (tasks T1 and T2); trend A6.
14. Corrective action	Plan gear inspection and replacement before alarm; shorten the oil sampling interval.
15. Expected post-maintenance telemetry	Features return to baseline only after gear replacement.
16. Verification result	Illustrative result: replacement planned at the next stop; A6 trend monitored until then.

5.6 C15 Post-repair verification after gearbox replacement [E][I]

Table 20. C15 case record

Element	Content
1. Case ID	C15. Illustrative return of axis 4 to production after replacement of the gear set.
2. Machine operating condition	Warm-up complete, program unchanged, payload unchanged.
3. Sensor reading / time-series pattern	20 cycles: M 13.04 Nm, per-cycle peak mean 21.8 Nm, minimum mean 4.3 Nm, A6 0.10 Nm, sigma 0.46 Nm, TCP 0.10 mm. During the first 3 cycles of run-in M was 13.9 Nm.
4. Baseline comparison	Within the healthy tolerance on every feature.
5. Threshold comparison	Below all thresholds; no sample at or above 23.1 Nm.
6. Relevant operating state	Production after verification.
7. Symptoms	None.
8. Candidate faults	Residual fault (F02, F03, F08) after the repair; verified repair.
9. Evidence supporting each candidate	All Document 03 verification criteria are met; the run-in transient decays within 3 cycles.
10. Evidence against alternatives	A residual friction fault would keep M above 13.5 Nm and a residual wear would leave A6 above 0.2 Nm.
11. Diagnostic reasoning	Every verification criterion is inside its band over 20 cycles, and the status stays normal for 600 s.
12. Final diagnosis	Verified repair; the baseline is stored with the new gearbox serial and program ID.
13. Recommended inspection	No further inspection; confirm after 24 h of production.
14. Corrective action	None.
15. Expected post-maintenance telemetry	M 12.5 to 13.5 Nm; A6 at most 0.2; sigma 0.40 to 0.55; peak at most 23.1; TCP at most 0.15 mm.
16. Verification result	Illustrative pass: 600 s soak normal and the 24 h recheck within band.

6. Comparison Tables

6.1 Telemetry signatures

Table 21. Telemetry signatures by case

Case	M (Nm)	Peak (Nm)	A6 (Nm)	Sigma (Nm)	TCP (mm)	Idle offset	Spikes / bumps	Onset
C01 (end of ramp)	23.2	31.1	2.9	1.0	0.65	n/a	none	ramp
C02	14.6	22.9	0.62	0.55	0.19	n/a	none	held
C03	19.3	27.1	1.70	0.82	0.43	n/a	none	held
C04	13.8 after	41.3 spike	0.14	0.52	0.35	0	spike	step
C05	13.3	27 occasional	0.35	0.50	0.21	0	at 2.5 s and 6.0 s	drift
C06	15.8	26.4	0.12	0.47	0.10	0	none	step
C07	14.4 to 13.2	22.5	0.10	0.47	0.10	0	none	start
C08	13.0 flat	13.0	0	0	0.10	n/a	none	step
C09	13 to 18 (gaps)	24.8 observed	unreliable	-	0.4	-	gaps	-
C10	17.0	24.4	0.12	0.52	0.11	+1.1	none	drift
C11	16.2	24	0.10	0.47	0.12	+3.2	none	step
C12	13.6	22 plus bump	0.20	0.46	0.10	0	bump 1.2 s	step
C16	24.1	35.2	3.34	1.13	0.66	n/a	one tail sample	held

6.2 Threshold interpretation

Table 22. Threshold interpretation by case

Case	Platform status	Feature level	Why the two differ	Action driven by
C01	normal then warn then alarm	L1 at t 16 s	Features lead the platform by about 40 s	Features
C02	normal	L1	Platform limits not reached	Features
C03	warn flicker	L2	Peak envelope persistent	Envelope
C04	alarm single sample	L4	Event plus TCP step	Event
C05	normal	L1	Spikes below limits	Phase pattern
C06	warn flicker	not a fault	Program change	Context
C07	normal	transient	Warm-up	State gate
C08	normal	LS	Signal flat	Data path
C09	normal	LS	Samples missing	Data path
C10	normal	L1 to L2	Friction	Lubricant
C11	normal	L2	Brake	Brake
C12	normal	L1	Local	Dress pack
C16	alarm	L4	All critical criteria	Stop

6.3 Candidate fault comparison

Table 23. Candidate fault comparison across cases (scores from the Document 03 evidence matrix)

Case	Leading	Score	Second	Score	Margin	Confidence	Decisive evidence
C01 (t = 64 s)	F01	14	F04	8	6	High	A6 and TCP ratios
C02	F01	14	F04	8	6	Medium to high (baseline assumed)	A6 and TCP ratios
C03	F01	14	F04	8	6	High	Same ratios
C04	F07	14	F05	6	8	High for the event	Isolated spike, TCP step, event log
C05	F05	10	F01 / F04	4	6	Medium to high	Phase-locked spikes, TCP large for dM
C06	F10	13	F07	11	2	Medium until tool data checked	Program log, swing change
C07	State gate	-	-	-	-	-	Time since start, decay
C08	F11	12	F07	7	5	High	Zero variance, other channels normal
C09	F12	-	F01	-	-	Medium	Gaps, reconnects, rssi
C10	F02 / F03 / F09	7 each	F01	6	0	Low until tests	Lubrication record, warm-up, brake
C11	F03	12	F08	8	4	Medium to high	Idle offset equal to M shift, hot brake
C12	F06	7	F07	5	2	Medium	Bump in one phase window
C13 W-A	F01	14	F04	8	6	High	Ratios
C13 W-B	F02 / F03 / F09	7 each	F01	6	0	Low	Warm-up and lubricant
C14	F01	12	F04	6	6	Medium	A6 appears after friction removed
C16	F01	14	F04	8	6	High	Locked ratios, TCP

6.4 Diagnostic evidence summary

Table 24. Diagnostic evidence by fault across cases

Fault	Evidence that confirmed it	Evidence that excluded it	Cases
F01	A6 / dM 0.2 to 0.4; (kN - 1) / dM near 0.14; dTCP / dM near 0.05; swing unchanged; ramp	Absent A6; idle offset; step onset	C01, C02, C03, C13, C14, C16
F02	Mean rise without ripple; warm-up offset; overdue lubrication	A6 present; metal in oil	C10, C13 W-B
F03	Idle offset equal to mean shift; brake warm	Brake cool	C11
F05	Phase-locked spikes; TCP high for small dM	Mean shift above 3 Nm	C05
F06	Bump in one phase window	Whole-cycle shift	C12
F07	Spike plus TCP step; swing change	Locked ratios	C04
F10	Step with swing change at program boundary	Ripple, TCP rise	C06
F11	sigma 0; other channels normal	Cycle present	C08
F12	Gaps, reconnects, rssi	Complete data	C09

6.5 Recommended actions

Table 25. Recommended actions by case

Case	First action	Second action	Time target [I]	Escalation
C01	Watch at L1; inspect oil and backlash at L2	Replace gearbox at L3	L2: next stop	L3 or L4
C02	Trend; lubricant sample	Relubricate if due	Next inspection	If M above 15
C03	Inspect at next stop	Plan replacement	Next stop	Alarm
C04	Stop; inspect	Re-master	Immediate	Specialist
C05	Retorque; backlash	Replace coupling	Next stop	If spikes grow
C06	Check program log	Re-baseline	Same shift	None
C07	None	Exclude warm-up	-	-
C08	Data path	Fix interface	Same shift	Controls
C09	Network	Restart	Same shift	IT
C10	Lubricant	Seal check	Next stop	If particles
C11	Brake	Drive	Next stop	If hot
C12	Dress pack	-	Next stop	-
C16	Stop	Replace gearbox	Immediate	Specialist

6.6 Verification criteria

Table 26. Verification criteria and results by case

Case	Pass criteria	Result in the case
C01 to C03, C16	M 12.5 to 13.5; A6 at most 0.2; peak at most 23.1; TCP at most 0.15; status normal 600 s	Reached after NORMAL command (simulated)
C04	No spike 600 s; TCP restored; backlash within OEM	Illustrative pass
C05	No spikes in 20 cycles	Illustrative pass
C06	New baseline stable	Illustrative pass
C08	Sigma 0.40 to 0.55; periodic 8 s	Illustrative pass
C09	No gaps over 5 minutes	Illustrative pass
C10	Warm-up offset at most 0.5	0.3 Nm
C11	Idle offset zero	Zero
C12	Residual below 1 Nm	Pass
C15	All criteria	Pass

7. Summary, Safety Notes, References and Revision

7.1 Diagnostic summary

- Cases C01 to C03 and C16 show that the project fault is recognisable from features about 40 s before the first platform warning at the default ramp, and that the three locked ratios separate it from friction, event and data faults.
- Cases C04 to C15 show how events, non-fault states, sensor faults and data loss produce thresholds crossings or normal status that can mislead a threshold-only agent.

7.2 Key decision criteria

- warn 26 Nm, alarm 30 Nm [P]; critical 34 Nm [I]; M band 12.5 to 13.5 Nm; A6 watch 0.4 Nm; ratios A6 / dM 0.2 to 0.4, (kN - 1) / dM about 0.14, dTCP / dM about 0.052 mm per Nm.

7.3 Safety and escalation notes

All inspections require lock-out and tag-out and the OEM procedure. L4 conditions and suspected collisions require a safe stop and specialist inspection. The cases are illustrations for a prototype and not plant history.

7.4 References and engineering basis

- Repository simulation equations and tests [P]; Documents 01 to 04 of this suite; general condition monitoring practice [E].

7.5 Revision notes

Table 27. Revision history

Rev	Date	Change
1.0	2026-10-08	Initial release; cases to be replaced by real plant history when available.